WORKED SOLUTION

STEP II 2007, Question 10

Mechanics • Statics with Friction

STEP II 2007, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q10

The key idea

This is a statics problem with only one awkward ingredient: the position of the centre of mass. Once we know where G sits on the axis, taking moments about the contact point gets rid of both the normal reaction and the friction, and gives P directly. Horizontal resolution then says friction equals P in size, and the friction law $|F| \leq \mu R$ does the rest. The modulus appears because, depending on α, P may have to push or pull.

Solution

The model

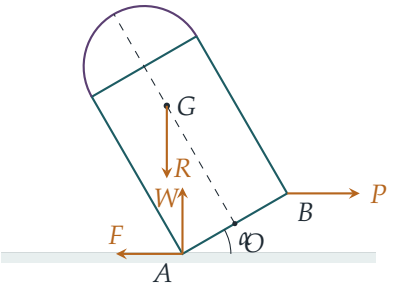
We treat the figure as a rigid body touching the plane at a single point A, the lowest point of the rim of the flat base (the free end of the cylinder). Let B be the highest point of that base, so AB is a diameter of length $2r$ making angle α with the horizontal. The forces are: the weight W at the centre of mass G; the normal reaction R and the friction F at A; and the horizontal force P at B.

The centre of mass

Measure distances along the axis from the centre O of the base. The cylinder has volume $3\pi r^3$, so mass $3\pi r^3\rho$, with centre of mass at $\frac{3}{2}r$. The hemisphere has volume $\frac{2}{3}\pi r^3$ and density 3ρ, so mass $2\pi r^3\rho$; its centre of mass is $\frac{3}{8}r$ beyond its flat face, which is at $3r$, so at $\frac{27}{8}r$. The total mass is $5\pi r^3\rho$, and

$$OG = \frac{3\cdot\frac{3}{2}r + 2\cdot\frac{27}{8}r}{5} = \frac{\frac{9}{2}r + \frac{27}{4}r}{5} = \frac{9}{4}r.$$

So G lies on the axis, $\frac{9}{4}r$ from the base, and $W = 5\pi r^3\rho g$.



Equilibrium

Take A as origin, with x horizontal (in the direction from A towards the foot of B) and y vertically up. Let P have horizontal component X in the x-direction, so X may be positive or negative; similarly let the friction be F in the x-direction.

The base runs from A in the direction $(\cos\alpha, \sin\alpha)$, so $B = (2r\cos\alpha, 2r\sin\alpha)$ and $O = (r\cos\alpha, r\sin\alpha)$. The axis is perpendicular to the base, pointing into the solid, in the direction $(-\sin\alpha, \cos\alpha)$. Hence the horizontal coordinate of G is

$$x_G = r\cos\alpha - \tfrac{9}{4}r\sin\alpha.$$

Resolving vertically, $R = W$. Resolving horizontally, $F + X = 0$, so $|F| = |X|$.

Ask yourself Which point should we take moments about?

About A: then R and F have no moment, leaving only W and P. The weight acts vertically through G, so its moment about A is Wx_G (clockwise when $x_G > 0$). The horizontal force at B acts at height $2r\sin\alpha$, so its moment is $X \cdot 2r\sin\alpha$ (anticlockwise when $X < 0$, clockwise when $X > 0$). Taking clockwise as positive,

$$Wx_G + 2r\sin\alpha\, X = 0 \quad\Longrightarrow\quad X = -\frac{W\left(r\cos\alpha - \frac{9}{4}r\sin\alpha\right)}{2r\sin\alpha} = W\left(\tfrac{9}{8} - \tfrac{1}{2}\cot\alpha\right),$$

dividing by $\sin\alpha$, which is positive since $0 < \alpha < \frac{1}{2}\pi$.

The friction condition

Since the plane is rough and the solid does not slip at A, we need $|F| \le \mu R$. Using $|F| = |X|$ and $R = W > 0$,

$$W\left|\tfrac{9}{8} - \tfrac{1}{2}\cot\alpha\right| \le \mu W \quad\Longrightarrow\quad \mu \ge \left|\tfrac{9}{8} - \tfrac{1}{2}\cot\alpha\right|,$$

which is the required result.

The sign of X tells us which way P acts. If $\tan\alpha > \frac{4}{9}$, then G lies beyond A on the side away from B, the figure would topple backwards on its own, and P must pull B away from the solid; if $\tan\alpha < \frac{4}{9}$, P must push. In either case friction opposes P, and the modulus covers both.

Checks

- If $\tan\alpha = \frac{4}{9}$, then $x_G = 0$: G is directly above A, the figure balances on its own, $P = 0$, and no friction is needed. The bound gives $\mu \geq |\frac{9}{8} - \frac{9}{8}| = 0$, as it should.
- At $\alpha = 45^\circ$ the bound is $\mu \geq \frac{5}{8}$.
- Numerically, slicing the solid into thin discs gives total mass $5\pi\rho r^3$ and $OG = 2.2500r$, and solving the three equilibrium equations directly for 10 000 random values of α gives $|F|/R$ equal to $|\frac{9}{8} - \frac{1}{2}\cot\alpha|$ every time.

Common slips

- Forgetting that the hemisphere has three times the density, and so taking its mass as $\frac{2}{3}\pi r^3\rho$; this moves G and changes the constant $\frac{9}{8}$.
- Measuring the hemisphere's centre of mass $\frac{3}{8}r$ from the base of the cylinder rather than from its own flat face at height $3r$.
- Assuming the direction of P (or of friction) and so losing the modulus: the result must hold whichever way P acts.

Beyond STEP

Finding centres of mass of composite bodies is the discrete shadow of a volume integral: $\bar{\mathbf{r}} = \int \rho\,\mathbf{r}\,\mathrm{d}V \big/ \int \rho\,\mathrm{d}V$, which is how the $\frac{3}{8}r$ for a hemisphere is derived in the first place. In first-year mechanics this grows into the theory of rigid bodies, where the same integrals with r^2 in place of $\mathbf{r}$ give moments of inertia. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both ideas carefully.

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